A NOTE ON VOTING

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Abstract. A framework is presented in which to analyze the power of players in a cooperative game in which only certain coalitions are allowed. The allowable coalitions are characterized by a closure operator that combinatorially abstracts the notion of convexity. The Shapley-Shubik index is extended to this situation, allowing a computation of the power of each player. One example of this framework, voting on a one-dimensional spectrum, is analyzed and it is shown that the powerful players are mid-way between the middle and the extremes. This framework is also applied to an empirical study of power on the Supreme Court. Implications and generalizations are also discussed.

I. Introduction.

Beginning with the paper of Shapley and Shubik [SS] cooperative game theory has been used to analyze the relative power of individuals in voting schemes. A common hypothesis in such analyses is that **all** coalitions of voters are possible, yet it is clear that this assumption is not realistic. Various structural and ideological issues will prevent some coalitions from forming.

Some attempts to deal with this problem were developed by incorporating coalition structures into the games [Gr]. These models do not restrict the possible coalitions so much as reduce the value of the coalitions that do not fit into the coalitional structure. When one is investigating the power of voters this is an important distinction, since a voter who moves from a low valued coalition to a highly valued one has considerable power, whereas if the first coalition is not permitted, that voter has no power.

More recently Faigle and Kern [FK] have investigated cooperative games under precedence constraints. Their idea is to restrict the allowable coalitions by using some underlying partially ordered set. They then present an axiomatization of the Shapley value in this restricted context and show that the Shapley value produced reflects the expected marginal contribution of a player to an allowable random coalition. Further work along these lines has been done by Gilboa and Lehrer [GL] and Bilbao [Bi3].

In this note we will restrict the allowable coalitions by using a combinatorial theory of convexity called *convex geometries*, a notion developed to combinatorially

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abstract geometric convexity. Convex geometries have appeared previously in the social science literature [Mo] and have been used in the theory of games by Bilbao [Bi1][Bi2] to extend the work of Owen [Ow] and Borm, et. al. [BO] to graph-restricted games. Bilbao uses a special case of convex geometries (so-called partition convex geometries) to develop local coalitional structures.

Much of the theory that we will need is implicit in the work of Faigle and Kern [FK] mentioned previously. The games under precedence constraints that they study are special cases of the games that examined in this paper, and much of what they prove goes through in the more general context. The more general notion of a convex geometry is more appropriate to describe voting. In particular the example in §III is not an example of a game under precedence constraints.

In the following section convex geometries are defined and some of their properties described. The definition of the Shapley-Shubik index is extended to this more general model. Section III is devoted to studying one model that fits this framework, voting in a one-dimensional spectrum. The values in this model are explicitly computed and the results analyzed. The results from this one-dimensional model seem to be in some conflict with the median voter hypothesis. In §IV this framework will be employed to do an empirical study of voting in the Supreme Court during the 1977-78 term, a term previously analyzed from a game-theoretic perspective. Finally, some remarks about future work will be made.

II. The Framework.

In this section the general framework of convex geometries will be discussed and then applied to the analysis of voting. Those readers who would like to see a specific application can skip this section and read §III where one such example is analyzed.

I begin with an axiomatization of allowable coalitions. If not all coalitions are allowed to form, what properties should the allowable ones have? I propose two axioms:

(A1) *(Intersection closed)* If A and B are allowable coalitions then $A \cap B$ is as well.

(A2) *(One-voter extension)* If A is an allowable coalition, not the grand coalition, then there is a voter x not in A so that $A \cup \{x\}$ is allowable.

The intersection closed axiom would seem to be intuitive. If one thinks of allowable coalitions of voters to be coalitions that agree on certain propositions, then intersections of allowable coalitions should also be allowable, since they agree on a larger set of propositions. The one-voter extension axiom is perhaps less obvious. The idea is that the number of issues to be decided by the voting system is large enough that allowable coalitions can be extended one voter at a time. From another perspective, this axiom says that the voters are independent enough that they do not ever have to act simultaneously to enlarge a coalition.

The reader should note that it is not required that the complement of an allowable coalition also be allowable. This may seem odd, since commonly one views voting as choosing between two alternatives and those voters who don't vote for one must vote for the other. Below it will be argued that voting is frequently more complicated than this and so requiring that allowable coalitions be closed under complementation is not required. There is also a technical explanation for us not requiring this:

Theorem 2.1. *Let X be a finite set and $\mathcal{L}$ be a collection of subsets of X satisfying axioms* **(A1)** *and* **(A2)**. *If in addition $\mathcal{L}$ has the property that $A \in \mathcal{L}$ implies that $X - A \in \mathcal{L}$ for all $A \subseteq X$, then $\mathcal{L}$ consists of all of the subsets of X.*

Proof. See Appendix I. □

Theorem 2.1 shows that the only collection of subsets that satisfies **(A1)** and **(A2)** and is closed under complementation is the collection of all subsets, i.e., every subset is allowable.

Here, then, are the formal definitions. Let X be a finite set of cardinality n and $\mathcal{L}$ be a collection of subsets of X with the properties

(1) $\varnothing \in \mathcal{L}$,
(2) $\mathcal{L}$ is intersection closed,
(3) $\mathcal{L}$ satisfies one-voter extension.

Call $\mathcal{L}$ a *convex geometry* on X. Convex geometries are a combinatorial abstraction of convex sets. The reader should see [EJ] for a discussion of their properties and a myriad of examples.

Alternatively one can think of $\mathcal{L}$ as being a *closure operator*, i.e., for any subset $A \subseteq X$ define the *closure of* A, $\mathcal{L}(A)$ to be

$$\mathcal{L}(A) = \bigcap_{\{C \in \mathcal{L} | C \supseteq A\}} C.$$

It is easy to check that $\mathcal{L}$ is a closure operator on X, i.e., $\mathcal{L}$ is a function from 2^X, the set of all subsets of X, to itself satisfying

(C1) $A \subseteq \mathcal{L}(A)$,
(C2) $A \subseteq B$ implies that $\mathcal{L}(A) \subseteq \mathcal{L}(B)$,
(C3) $\mathcal{L}(\mathcal{L}(A)) = \mathcal{L}(A)$.

with the additional condition that $\mathcal{L}(\varnothing) = \varnothing$. The subsets in $\mathcal{L}$, or equivalently those subsets of X of the form $\mathcal{L}(A)$ for some $A \subseteq X$ are usually called *convex sets*. For us, the convex sets are the allowable coalitions in the game.

If $A \in \mathcal{L}$ and $x \in A$, then call x an *extreme point* of A if $A - x \in \mathcal{L}$. The set of extreme points of A will be denoted $ex(A)$. The extreme points of an allowable coalition are those voters whose removal leaves an allowable coalition.

Theorem 2.2. *A convex geometry $\mathcal{L}$ on a set X has the property that for every $A \in \mathcal{L}$, $A = \mathcal{L}(ex(A))$.*

Proof. See [EJ,Theorem 2.1] □

Before defining the Shapley-Shubik index the notion of a *maximal chain* in $\mathcal{L}$ must be discussed. A *maximal chain* M of $\mathcal{L}$ is an ordered collection of convex sets

$$M = (\varnothing = M_0 \subset M_1 \subset \cdots \subset M_{n-1} \subset M_n = X)$$

so that there does not exist a convex set C and an index j, $0 \leq j \leq n-1$ such that $M_j \subsetneq C \subsetneq M_{j+1}$. That is, M is a chain of convex sets that is not contained in any larger chain. From the one-voter extension axiom and induction one can prove the following

Theorem 2.3. *A collection $\mathcal{L}$ of subsets of X is a convex geometry if and only if $\mathcal{L}$ is intersection closed, $\varnothing \in \mathcal{L}$, and every maximal chain contains $|X|+1$ convex sets. In particular, if*

$$M = (\varnothing = M_0 \subset M_1 \subset \cdots \subset M_{n-1} \subset M_n = X)$$

then $n = |X|$ and $|M_i| = i$ for all $0 \le i \le n$. $\square$

Proof. See [EJ,Theorem 2.2] $\square$

Note that Theorem 2.3 implies that in every maximal chain M of a convex geometry $\mathcal{L}$, there is some minimal convex set $M^x \in M$ that contains x. Moreover, $x \in ex(M^x)$ and hence $M^x - x \in \mathcal{L}$.

Example. Consider the convex geometry

$$\mathcal{L} = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}.$$

One can draw a picture of $\mathcal{L}$ called the *Hasse diagram* as follows

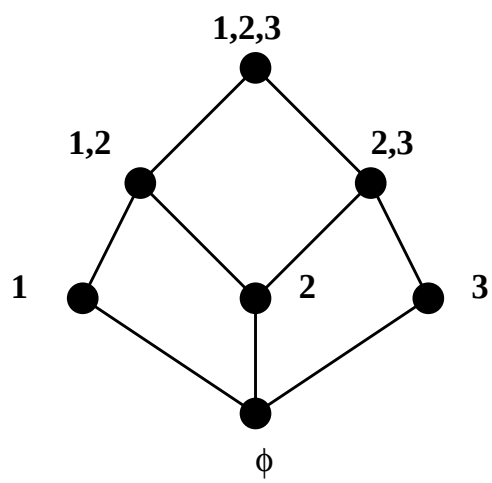

FIGURE 1

where the edges in the diagram represent adding an element to the bottom set to get the top set. The maximal chains in $\mathcal{L}$ correspond to paths from $\varnothing$ to $\{1,2,3\}$ that go along the edges in the diagram always going up. There are four such chains:

$$\varnothing \subset \{1\} \subset \{1,2\} \subset \{1,2,3\}$$
$$\varnothing \subset \{2\} \subset \{1,2\} \subset \{1,2,3\}$$
$$\varnothing \subset \{2\} \subset \{2,3\} \subset \{1,2,3\}$$
$$\varnothing \subset \{3\} \subset \{2,3\} \subset \{1,2,3\}$$

The next step is to consider a function that assigns a value to each coalition. In voting models such a function is usually 0 or 1 depending on whether that coalition can win the election. The actual value does not matter for the definition. Consider

a function $v : \mathcal{L} \to \mathbb{R}$ that assigns a value to each convex set in $\mathcal{L}$. For each $x \in X$, define the *Shapley-Shubik index of* x, $SS(x)$, to be the sum

$$SS(x) = \frac{1}{m} \sum_{M} v(M^x) - v(M^x - x)$$

where the sum is taken over all maximal chains of $\mathcal{L}$ and m is the total number of maximal chains of $\mathcal{L}$. It is trivial to check that

$$\sum_{x \in X} SS(x) = v(X) - v(\varnothing).$$

The function SS measures the average marginal contribution of a voter x to a random allowable coalition. One can give an axiomatic characterization of this function as well by following the work of [FK,§2]. This will be done elsewhere [BE]

In the case that $\mathcal{L} = 2^X$, that is when every subset of X is allowable, the above definition specializes to the usual definition of the Shapley-Shubik index[St]. An analogous definition can be made for the Banzhaf index [Ba][St]. Such a definition will be discussed in the last section of this paper.

III. An example: voting in a one-dimensional spectrum.

Imagine the voters in an election distributed along a line. Assume that the allowable coalitions are intervals along this line, that is coalitions can form in the middle against the outside but not vice versa. Where is the power in such an election?

To model this situation, let $X = \{1, 2, \ldots, n\}$ for some positive integer n, and let $\mathcal{L}_n$ be the collection of sets of the form $[i, j] = \{i, i+1, \ldots, j-1, j\}$ for $1 \leq i \leq j \leq n$. When $n = 2k + 1$ is odd let v be the "majority function", i.e.,

$$v(A) = \begin{cases} 1 \ if \ |A| \geq k+1 \\ 0 \ if \ |A| \leq k \end{cases}$$

for all $A \in \mathcal{L}$.

Example. For $k = 1$,

$$\mathcal{L}_3 = \{\varnothing, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 2, 3\}\},$$

the same convex geometry that was discussed earlier. The function v assigns 1 to the subsets $\{1, 2\}, \{2, 3\}, \{1, 2, 3\}$ and 0 to the rest. Since the maximal chains in this convex geometry have already been listed above, it is easy to compute SS from the definition, e.g.,:

$$\begin{aligned} SS(1) &= \frac{1}{4}[(v(\{1\}) - v(\varnothing)) + (v(\{1, 2\}) - v(\{2\})) \\ &\quad + (v(\{1, 2, 3\}) - v(\{2, 3\})) + (v(\{1, 2, 3\}) - v(\{2, 3\})] \\ &= \frac{1}{4}[0 + 1 + 0 + 0] \\ &= \frac{1}{4}. \end{aligned}$$

An elementary combinatorial analysis shows that

Theorem 3.1. *If v is the majority function on the convex geometry $\mathcal{L}_{2k+1}$ then*

$$SS(i) = \begin{cases} \frac{\binom{k}{i-1}}{2^{k+1}} \; if 1 \leq i \leq k \\ \frac{1}{2^k} \; if \; i = k+1 \\ \frac{\binom{k}{2k+1-i}}{2^{k+1}} \; if \; k+2 \leq i \leq 2k+1 \end{cases}$$

Proof. A proof is sketched in Appendix II. □

The following remarks should be made about the model:

(1) The only voters less powerful than the median voter are the two extreme voters (if $k \geq 3$). In an election of 101 voters, $SS(1) \approx 4 * 10^{-16}$ and $SS(51) \approx 8 * 10^{-16}$.

(2) The most powerful voters are midway between the extremes and the median. Their power is enormous. For example, in an election with 101 voters, $SS(26) = .056$ which is five times what the usual Shapley-Shubik analysis would give and 10^{14} times the power of the median voter. In fact, for large k, $SS(\frac{k}{2}+1) \approx \frac{1}{\sqrt{2\pi k}}$.

The above observations are illustrated in Figure 2, where a graph of the function SS in the situation where $k = 50$ is shown.

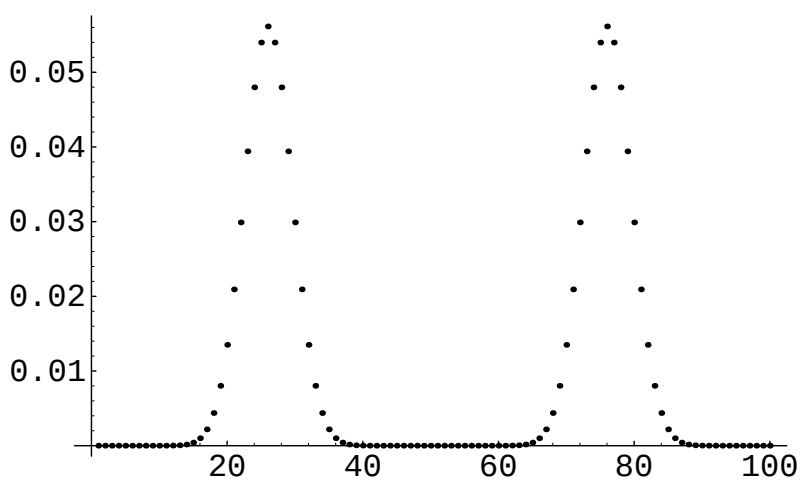

FIGURE 2

(3) This model would seem to be at odds with the median voter hypothesis. See, for example, [Mu,page 65]. Questions about the median voter hypothesis have been raised, primarily because its predictions are not confirmed by the data. This has given rise to a number of theoretical attacks as well [IV][AR].

(4) There is another voting power index that applies in this example, the *attitudinal dependent*, *AD*, index of Shapley [Sh],[FS]. A similar index was described by Owen[Ow2]. See also [OS]. In this one-dimensional model the *AD* index assigns all of the power to the median voter and none to any

other, thus strongly confirming the median voter hypothesis. The difference between the AD index and the one in this paper consists of which coalitions are allowable. In the terminology from above, the allowable coalitions for the AD index are all of the form $\{1, 2, \ldots, i\}$ or $\{j, j+1, \ldots, n\}$ for all $1 \leq i, j \leq n$. Thus, the AD index looks at allowable coalitions that satisfy one-voter extension and are closed under complementation. Notice that the set of allowable coalitions is not intersection closed. Note also that the AD index can not be used to model votes in a one-dimensional spectrum if some voters at both extremes do not vote at all. It is perhaps not a coincidence that an election in which alienation is allowed can exhibit behavior different from that of the median voter hypothesis [Mu,10.A]. This is a severe limitation that makes the AD index unsatisfactory for the one-dimensional analysis. The AD index will be discussed further in the next section.

IV. A comparison with the AD index. In this section I will show how the framework of convex geometries and the SS index can be used as an alternative to the AD index of Shapley [Sh],[FS]. By way of an illustration I will re-examine the analysis of the the 1977-78 term of the Supreme Court done by Frank and Shapley [FS].

Recall that the AD index is designed to compute a power index for a set of voters that are viewed as points in some Euclidean space. The basic idea is as follows. Let X be a finite set of $2k+1$ points in $\mathbb{R}^n$. A generic linear functional linearly orders the points based upon the value of the linear functional on each point. A value of 1 is assigned to the point that appears in the $k+1$ place of this order and 0 to all of the rest. The index is computed by integrating this function over all the linear functionals relative to a certain natural measure.

Example. Consider the set of 5 points in the plane in Figure 3. A particular linear functional L is indicated by the oriented line in the Figure. Relative to this functional the linear ordering produced is $3, 5, 4, 2, 1$.

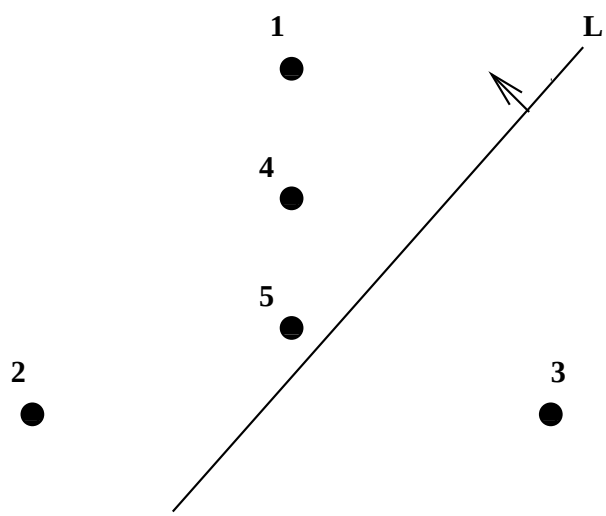

FIGURE 3

Note that a consequence of using linear functionals to define the linear ordering

is that the coalitions of voters that can form are those that can be separated from their complement by hyperplane. Thus in Figure 3, the coalition $\{4, 5\}$ can never materialize, nor can its complement $\{1, 2, 3\}$. In particular, a coalition can form if and only if its complement can form as well.

There is a standard way of producing a convex geometry from a collection of points in space. Let X be a collection of points in $\mathbb{R}^n$. Fix a subset A of X. We will say that a point $t \in \mathbb{R}^n$ is in the *convex hull of* A if t can be written in the form

$$t = \sum_{a \in A} \lambda_a \, a$$

where $\lambda_a \geq 0$ for all $a \in A$ and $\sum \lambda_a = 1$. We will denote the convex hull of A by $convA$.

Let $\mathcal{L}$ be the set of subsets of X defined by

$$\mathcal{L} = \{A \subseteq X \,|\, convA \cap X = A\}\,.$$

That is, the set of convex sets $\mathcal{L}$ are those subsets of X whose convex hull contains no additional points of X. We will call $\mathcal{L}$ the *Euclidean convex geometry* on X. It is elementary to see that $\mathcal{L}$ is a convex geometry on X [EJ,§3 Example 1].

Example. Let X be the five points in the plane in Figure 3. The point 4 is in $conv\{1, 2, 3\}$. The set $\{2, 3, 4\}$ is not in $\mathcal{L}$ since $5 \in conv\{2, 3, 4\}$ and hence

$$conv\{2, 3, 4\} \cap X \neq \{2, 3, 4\}.$$

Here then is description in words of the two different indices in question. For the AD index, linear orderings of the voters with the property that each initial set of voters can be separated from the rest by a hyperplane are considered and these orderings are weighted by the probability that a linear functional gives rise to that ordering. For the SS index of the Euclidean convex geometry, linear orderings of the voters with the property that the convex hull of each initial set of voters does not contain any of the later voters are considered and these orderings are given equal probability.

Frank and Shapley applied the AD index to analyze the voting power on the 1977-78 Supreme Court Term [FS]. By using a factor analysis they coordinatized each justice in $\mathbb{R}^3$ and then proceeded as above. The coordinatization that they use is shown in Appendix 3. The resulting index is:

Justice	**Brennan**	**Blackmun**	**Burger**	**Marshall**	**Powell**	**Rehnquist**	**Stevens**	**Stewart**	**White**
Index	**.033**	**.092**	**.065**	**.080**	**.346**	**.051**	**.047**	**.157**	**.129**

Frank and Shapley actually performed two different computations, based on two different transformations of their data. The table shown here corresponds to Transformation 1 in [FS,Table 3]. The results are very similar for Transformation 2.

In [EC] I argue that this index is implausible and provide a different empirical model that I believe gives better results. From a theoretical standpoint, as mentioned above, the requirement of the AD index that the coalitions be closed under complementation is too strong. In any voting in which there is more than two alternatives it is hard to justify requiring that the complement of an allowable coalition be allowable. It might appear that for the Supreme Court there are only two possibilities in voting on a particular case, to affirm or to reverse. On the other hand, what is more important from a legal standpoint is which justices concur in an opinion since it is the opinions that set legal precedent. Thus while two justices may agree on the outcome of a case they may do so for vastly different reasons, and to identify them as being in a coalition would be to ignore these very important differences. Thus the Court fractures in to possibly many different opinions and the coalitions formed are not closed under complementation.

Suppose instead we compute the SS index using the Euclidean convex geometry. The resulting index is:

Justice	Brennan	Blackmun	Burger	Marshall	Powell	Rehnquist	Stevens	Stewart	White
Index	.111	.094	.105	.111	.142	.108	.111	.108	.111

This index would seem to be more plausible than that of Frank and Shapley in that it does not exhibit the wide discrepancies in power between justices who seem to be similar [EC]. It also is consistent with the results of a different empirical approach discussed in [EC].

V. Further Remarks.

Remark 1. As mentioned in §II, there is an analogous way to extend the definition of the Banzhaf index $B(x)$ to this class of models namely

$$B(x) = \frac{1}{r} \sum_{\{C \in \mathcal{L} \ | \ x \in ex(C)\}} v(C) - v(C - x).$$

where r is taken to normalize the sum so that $\sum_{x \in X} B(x) = 1$. When applied to the example in §III the following index results:

$$B(i) = \begin{cases} \dfrac{1}{2k+2} \ if \ i \neq k+1 \\ \dfrac{1}{k+1} \ if \ i = k+1 \end{cases}$$

This is an unsatisfactory answer since it allocates the same power to all voters except the median. As of yet, there is no theoretical framework to explain *a priori* a preference for the Shapley-Shubik index, but this example motivates the choice.

Remark 2. A natural next step would be to explore a two dimensional model of voting. This is more complex than it may seem, and it is doubtful that a complete solution of the sort found in §III can be found. On the other hand, numerical simulations could be done and might prove enlightening.

Remark 3. The one dimensional model described in the earlier section makes some very strong predictions that one might be able to check empirically. That work will be left for the future.

Appendix I: The proof of Theorem 2.1. Let us suppose that $X = \{1, 2, \ldots, n\}$ and let 2^X denote the collection of all subsets of X. Suppose that every subset $X - i,\ 1 \leq i \leq n$ is in $\mathcal{L}$. Since every subset $A \subseteq X$ can be written as

$$A = \bigcap_{i \notin A} X - i$$

and since $\mathcal{L}$ is intersection closed one can conclude that $\mathcal{L} = 2^X$ which would finish the proof.

Thus assume that some of the $X - i$'s are not in $\mathcal{L}$. Since $\mathcal{L}$ is closed under complementation, if $X - i \notin \mathcal{L}$ it follows that $i \notin \mathcal{L}$ as well. Let $J = \{i \in X \mid i \notin \mathcal{L}\}$. I claim that $J \in \mathcal{L}$ since

$$J = \bigcap_{i \in \mathcal{L}} X - i.$$

Take

$$M = (\varnothing = M_0 \subset M_1 \subset \cdots \subset M_{n-1} \subset M_n = X)$$

to be any maximal chain in $\mathcal{L}$ and consider the collection of sets

$$\varnothing = M_0 \cap J \subseteq M_1 \cap J \subseteq \cdots \subseteq M_{n-1} \cap J \subseteq M_n \cap J = J.$$

Since M was a maximal chain it is known that $|M_i| = i$, and thus it follows that $|M_i \cap J| - |M_{i-1} \cap J|$ is either 0 or 1. Since $|M_0 \cap J| = 0$ and $|M_n \cap J| = |J| \geq 1|$ one sees that one of the terms $M_i \cap J$ must have cardinality 1, i.e., $M_i \cap J = k$ for some $k \in X$. Since $M_i \cap J = k$ and J and M_i are in $\mathcal{L}$, we know that $k \in \mathcal{L}$. But $k \in J$ and so $k \notin \mathcal{L}$ by construction. Thus there is a contradiction and hence $J = \varnothing$ and $\mathcal{L} = 2^X$, which was to be proven. $\square$

Appendix II: The proof of Theorem 3.1. In this appendix I sketch a proof for Theorem 3.1. Before the start of the proof the notion of chains in $\mathcal{L}$ must be refined. Fix $C, D \in \mathcal{L}$ where $C \subset D$. A *saturated chain* from C to D is an ordered collection of convex sets

$$C = C_0 \subsetneq C_1 \subsetneq \cdots \subsetneq C_t = D$$

such that C_i differs from C_{i-1} by the addition of exactly one element, i.e., $|C_i| = |C_{i-1} + 1|$ for $1 \leq i \leq t$. It follows that $t = |D| - |C|$. Note that maximal chains are the saturated chains from $\varnothing$ to X.

The proof begins with a few lemmas.

Lemma 1. *The number of maximal chains in $\mathcal{L}_n$ is 2^{n-1}.*

Proof. The proof is by induction. For n equal to 1 or 2 the lemma is clearly true. For $n \geq 3$ note that every maximal chain has as its second to last set either $[1, n-1]$ or $[2, n]$. The number of maximal chains that go through $[1, n-1]$ is clearly the same as the number of saturated chains from $\varnothing$ to $[1, n-1]$ which is equal to the number of maximal chains in $\mathcal{L}_{n-1}$ and hence, by induction, there are 2^{n-2} of them. Similarly the number of maximal chains through $[2, n]$ is the same as the number of maximal chain in $\mathcal{L}_{n-1}$ and so there are 2^{n-2} of those. Thus, the total number of maximal chains in $\mathcal{L}_n$ is $2^{n-2} + 2^{n-2} = 2^{n-1}$. $\square$

Lemma 2. *If $C = [i, j]$ then the number of saturated chains from $\varnothing$ to C is 2^{j-i} and the number of saturated chains from C to $[1, n]$ is $\binom{n-j+i-1}{i-1}$.*

Proof. The saturated chains from $\varnothing$ to C are clearly in one-to-one correspondence with the maximal chains in $\mathcal{L}_{j-i+1}$ and so there are 2^{j-i} of them.

The convex sets $D \supseteq C$ are all of the form $C + [t, i-1] + [j+1, s]$. So to build a saturated chain from C to $[1, n]$ one unions the terms $[1, i-1]$ in reverse order interleaved with the terms $[j+1, n]$ in order. One can think of this process as a permutation of the set $[1, i-1] \cup [j+1, n]$ in which the numbers $[1, i-1]$ appear in reverse order and the numbers $[j+1, n]$ appear in order. This is a permutation of $n - i + j - 1$ symbols and if the **positions** of the numbers $[1, i-1]$ are known then the permutation can be uniquely reconstructed. There are $\binom{n-j+i-1}{i-1}$ ways of choosing those positions, so the lemma is proven. $\square$

Proof of Theorem 3.1. Fix i such that $1 \leq i \leq k$. Then $C = [i, k+i]$ is the unique convex set in $\mathcal{L}_{2k+1}$ such that

(1) $i \in ex(C)$,
(2) $v(C) = 1$, and
(3) $v(C - i) = 0$.

So the only maximal chains in $\mathcal{L}_{2k+1}$ that will contribute to the value of $SS(i)$ are those chains that contain $C - i$ and C, and each of those chains contributes one to the sum. The number of such chains is equal to the product of the number of saturated chains from $\varnothing$ to $C - i$ times the number of saturated chains from C to $[1, 2k+1]$. These numbers are 2^{k-1} and $\binom{2k+1-(k+i)+i-1}{i-1} = \binom{k}{i-1}$, respectively, by Lemma 2. The total number of maximal chains in $\mathcal{L}_{2k+1}$ is 2^{2k} and thus

$$SS(i) = \frac{1}{2^{2k}} 2^{k-1} \binom{k}{i-1} = \frac{1}{2^{k+1}} \binom{k}{i-1}.$$

The formula for $SS(i)$ when $k + 2 \leq i \leq 2k + 1$ follows from symmetry since $SS(i) = SS(2k + 2 - i)$.

If $i = k + 1$ then there are two types of contributors to the sum $SS(k + 1)$, maximal chains containing both $[k+2, 2k+1]$ and $[k+1, 2k+1]$ and maximal chains containing both $[1, k]$ and $[1, k+1]$. The result follows from a similar analysis to that above. $\square$

Appendix III. The Frank-Shapley model. Using a factor analysis Frank and Shapley assigned the following coordinates to the justices of the 1977-78 Supreme Court in a three-dimensional space:

Burger	1.074	.407	−.384
Brennan	−1.433	−.934	.723
Stewart	−.011	−.145	−.206
White	−.424	1.173	.899
Marshall	−1.262	−.916	.542
Blackmun	−.015	1.415	−.040
Powell	.034	.947	−.001
Rehnquist	1.774	−1.142	1.266
Stevens	.021	−.820	1.694

For various reasons they then proceeded to manipulate the coordinates using a linear transformation. Applying a linear transformation to the points leaves the structure of the convex sets invariant, so I used these points as the ones for computing the SS index. The convex hull of a set of points was computed using the program `qhull`.

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